

Operation Protocol

Open-Source Cable-Actuated Snake Rescue Tool with Rotation-Assisted Body Winding

For dummy-snake/rubber-tube prototype testing and trained-handler evaluation

Hardware name	Open-source cable-actuated snake rescue tool with rotation-assisted body-winding restraint mechanism and compliant soft-grip jaw interface
Document type	Operation protocol
Intended use	Prototype operation, dummy-snake testing, training demonstration, and controlled evaluation by trained personnel
Validation condition	Dummy snake / flexible rubber tube only
Live-animal use	Restricted to trained snake rescuers or authorized wildlife personnel under applicable ethical approval and local wildlife regulations

Safety notice: This tool is an early-stage open-source hardware prototype. The protocol below is intended for dummy-snake/rubber-tube functional testing and controlled demonstration. It is not a field-certified live-snake rescue procedure. Live-snake operation should only be performed by trained and authorized personnel with appropriate safety controls, local permissions, and ethical approval.

1. Purpose

This operation protocol describes the recommended functional sequence for using the cable-actuated snake rescue tool during prototype testing. The protocol explains initial body engagement, secure body capture, manual working-end rotation, body winding around the gripper/shaft region, controlled head-restraint positioning, and spring-assisted release.

2. Required equipment

- Assembled snake rescue tool with cable-actuated jaw, spring-return linkage, soft-grip jaw pads, and manually rotatable working-end assembly.
- Flexible dummy snake or rubber tube for prototype validation.
- Secure transport container or bucket for release/transfer demonstration.
- Personal protective equipment for trained evaluators, if required by the testing environment.
- Basic inspection tools, such as vernier caliper, spring scale/digital force gauge, and stopwatch, if measurements are being recorded.

3. Pre-operation inspection

- Confirm that all fasteners, collars, brackets, jaw pivots, linkage arms, and cable guides are tightened and aligned.
- Check that the brake-style trigger moves smoothly and returns to its original position after release.
- Check that the cable is correctly routed through the guide supports without sharp bends, fraying, slack loops, or rubbing.
- Confirm that the spring-return jaw opens automatically when the trigger is released.
- Inspect the soft-grip jaw pads for secure attachment, wear, cracks, or detachment.
- Manually rotate the working-end assembly through the expected rotation range and confirm that the cable, spring, linkage arms, and jaw pads do not twist, lock, or interfere.

- Use only a dummy snake or flexible rubber tube during prototype-stage testing unless live-animal testing has been approved and is conducted by trained personnel.

4. Functional operation sequence

The intended operation sequence is summarized in Table 1. The same sequence should be used during dummy-snake functional testing and demonstration.

Stage	Procedure
Stage 1 - Approach	Hold the tool by the operator handle and approach the dummy snake from a safe distance. Position the working-end jaw assembly near a suitable body region rather than directly targeting the head.
Stage 2 - Initial body grip	Pull the brake-style trigger to tension the steel cable and close the soft-grip jaw pads around the dummy snake body.
Stage 3 - Secure body capture	Maintain trigger input until the dummy snake body is held between the compliant jaw pads. Confirm that the body is engaged without visible slipping or jaw-pad detachment.
Stage 4 - Working-end rotation	After the body is captured, manually rotate the working-end assembly about the shaft axis using the secondary rotating control/working-end structure.
Stage 5 - Body winding	Continue controlled rotation so that the captured dummy snake body rotates and winds around the gripper/shaft region. Observe whether winding occurs without cable, spring, or linkage interference.
Stage 6 - Controlled head-restraint positioning	As body winding progresses, the dummy snake head is brought into a controlled restrained position near the working-end structure. This is demonstrated only under dummy-snake testing conditions.
Stage 7 - Spring-assisted release	Position the tool over a secure container or release area. Release the brake-style trigger and confirm that the spring-return jaw opens and the dummy snake can be removed or deposited without sticking.

5. Detailed procedure

5.1 Approach and positioning

Place the dummy snake on a clear test surface. Hold the tool using the operator handle and align the working-end jaw assembly with a suitable body region. Avoid placing the jaw directly on the head during the initial approach.

5.2 Trigger actuation

Pull the brake-style trigger gradually. The cable should transmit the input to the front linkage and close the jaw. The soft-grip jaw pads should contact the body surface and hold it without rigid metal contact.

5.3 Body-capture confirmation

Verify that the dummy snake body is held between the jaw pads. The jaw should remain aligned, and the cable should not slip from the guide supports. If the dummy snake slips immediately, adjust cable tension or jaw-pad contact before repeating the test.

5.4 Working-end rotation

Rotate the working-end assembly manually through the intended range. During rotation, observe the cable, spring, linkage arms, and jaw pads. Stop the test if binding, cable twisting, linkage locking, or pad detachment occurs.

5.5 Body winding around the gripper/shaft region

Continue rotation until the dummy snake body winds around the gripper/shaft region. This stage demonstrates the main rotation-assisted body-winding function of the prototype.

5.6 Controlled head-restraint positioning

Observe whether body winding reduces uncontrolled movement and brings the dummy snake head into a controlled restrained position near the working-end structure. This result should be reported as dummy-snake prototype behavior, not live-snake performance.

5.7 Release and reset

Move the tool over a secure release area or container. Release the trigger and confirm that the spring-return mechanism opens the jaw. Remove or deposit the dummy snake and reset the tool for the next test cycle.

6. Optional measurement protocol

The following measurements may be recorded during prototype validation:

- Maximum jaw opening and minimum jaw gap after trigger pull.
- Soft pad contact width.
- Grip force using a spring scale or digital force gauge.
- Trigger travel and cable displacement at the working end.
- Maximum working-end rotation angle and practical body-winding range.
- Slip distance before and after 90°, 180°, and 270° working-end rotation.
- Release time after trigger release.
- Number of repeated trigger open-close cycles and working-end rotation cycles.

7. Functional acceptance criteria

Stage	Procedure
Jaw actuation	Jaw closes when trigger is pulled and reopens when trigger is released.
Cable routing	Cable remains aligned in the guide supports without slipping or excessive rubbing.
Spring return	Spring returns the movable jaw to the open position after trigger release.
Soft-grip contact	Jaw pads remain attached and contact the dummy snake body without rigid metal-to-body contact.
Working-end rotation	Working end rotates through the intended range without cable, spring, or linkage interference.
Body winding	Dummy snake body rotates and winds around the gripper/shaft region during working-end rotation.
Release	Dummy snake can be released without sticking, jaw locking, or pad detachment.

8. Safety limitations and warnings

- This protocol does not certify the tool for live venomous snake handling.
- The tool should not be used by untrained individuals for live-snake rescue.
- Validation with a dummy snake does not reproduce live-snake muscular resistance, defensive movement, surface friction, body stiffness, or strike behavior.
- Grip force depends on trigger input, cable tension, jaw geometry, and pad compliance; a calibrated force-limiting mechanism has not yet been integrated.

- Any future live-snake use must follow institutional ethics approval, local wildlife regulations, and supervision by trained snake-handling experts.

9. Post-operation inspection and maintenance

- Inspect the jaw pads for wear, detachment, or damage.
- Check the cable for fraying, loosening, or stretch.
- Inspect the spring, pivots, bolts, and linkage arms for deformation or loosening.
- Clean and disinfect contact surfaces if the tool is used in biological or field environments.
- Retighten fasteners and adjust cable tension before the next operation if looseness is observed.
- Record any failure, interference, or abnormal behavior in the validation log.

10. Version information

Protocol version	1.0
Associated manuscript	Open-Source Cable-Actuated Snake Rescue Tool with Rotation-Assisted Body Winding for Controlled Head Restraint
Repository	[Insert GitHub repository URL]
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End of operation protocol